

Antidiabetic activity of a xanthone compound, mangiferin

Toshihiro Miura¹, Hiroyuki Ichiki², Itsuko Hashimoto¹, Naoki Iwamoto¹, Motoshi Kato¹, Masayoshi Kubo², Eriko Ishihara¹, Yasuhiro Komatsu², Minoru Okada², Torao Ishida³ and Keiichiro Tanigawa¹

¹ Suzuka University of Medical Science, Suzuka, Mie, Japan

² Hi-tech Research Center, Suzuka University of Medical Science, Suzuka, Mie, Japan

³ Tsumura Central Research Laboratories, Tsumura and Co., Yoshiwara, Ami-machi, Inashiki-gun, Ibaraki, Japan

Summary

Mangiferin (MF) isolated from *Anemarrhena asphodeloides* Bunge rhizome, was tested for antidiabetic activity in KK-Ay mice, an animal model of type-2 diabetes. MF lowered the blood glucose level of KK-Ay mice 3 weeks after oral administration ($p < 0.01$). However, no effect on the blood glucose level in normal mice was seen, indicating that MF could be useful in treating type-2 diabetes. In addition, MF improved hyperinsulinemia and, on insulin tolerance test, reduced blood glucose levels of KK-Ay mice. From these findings, it seems likely that MF exerts its antidiabetic activity by decreasing insulin resistance.

Key words: mangiferin; KK-Ay mice; antidiabetic activity; insulin resistance

■ Introduction

Many natural components participate in a variety of antidiabetic functions *in vivo* (Kato et al., 1995, Miura et al., 1996, Kako et al., 1997). In an attempt to obtain effective new compounds to reduce blood glucose levels in diabetics, we isolated xanthone compounds associated with antidiabetic activity by bioassay-guided fractionation. In a previous study, we reported antidiabetic activity on single administration of mangiferin (Ichiki et al., 1998). There is, however, no evidence about antidiabetic activity on repeated administration and its mechanism. The aim of this study was, therefore, to examine the effect of the xanthone compound on glycemic metabolism in KK-Ay mice, an animal model of obese-type genetic type-2 diabetes with hyperinsulinemia.

■ Experimental Section

Mangiferin

Mangiferin (MF) (Figure 1) was isolated by the conventional method (Frahm and Chaudhuri, 1979, Fujita

and Inoue, 1982). The water extract of *Anemarrhena asphodeloides* Bunge rhizome (water 21, 2 h reflux) (5.5 g) was chromatographed on a diaion HP-20 column, using water (3.2 g), 50% methanol (1.1 g) and 100% methanol (1.0 g) as eluents. The 50% MeOH layer of AA (active layer) was analyzed by HPLC (MeCN : H₂O : 85% phosphoric buffer = 10 : 89 : 1, ODS column length 250-mm, diameter 46 mm, flow rate 0.8 ml/min) [retention time of Mangiferin (MF) = 36.8]. A voucher specimen was deposited at the herbarium of Kobe Pharmaceutical University (Her-

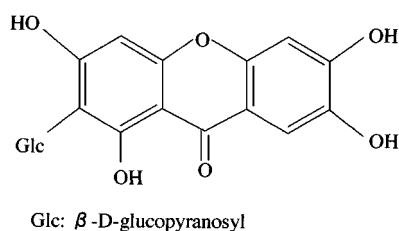


Fig. 1. Structure of Mangiferin (MF).

barium No. KPUH 22). The structure of MF was confirmed using spectroscopic methods (Frahm and Chaudhuri, 1979, Fujita and Inoue, 1982). The yield of mangiferin was 0.6%.

Antidiabetic Activity

Normal male mice (ddY, 12 weeks old) and KK-Ay mice (12 weeks old) were used. The mice were given free access to drinking water and food. Mangiferin was dissolved in distilled water and administered to the mice orally. As control, a distilled water solution was also administered to the mice. Blood samples were drawn from the eye with a capillary.

Insulin-Tolerance Test

Insulin-tolerance tests were performed at the end of the repeated study. Twenty-four hours before the test, food was removed from the cages. Insulin solution (0.5 U/kg body wt.) was administered subcutaneously. Blood samples were collected before the administration of the insulin and at 30, 60 and 120 minutes after.

Determination of Blood Glucose and Insulin

The glucose level of the drawn blood was determined using the glucose-oxidase method (Stevens, 1951) and serum insulin was measured by the double-antibody method (Bailei and Ahmed-Sorour, 1980).

Statistical Analysis

All data are expressed as mean $\pm$ SD. The student's *t* test was used for statistical analysis. Values were considered to be significantly different when the *p* value was less than 0.05.

Results and Discussion

Results in the KK-Ay mice are summarized in Figure 2. Mangiferin showed antidiabetic activities at a dosage of 30 mg/kg body wt. ($p < 0.01$). In normal mice, no change in blood glucose was observed as in MF-treated mice (data not shown). Thus it appears that MF could be useful in therapy for type-2 diabetes. The serum insulin concentration in MF-treated KK-Ay mice decreased (Figure 3). KK-Ay mice, including ob/ob mice and KK mice, with genetically induced diabetes are hyperinsulinemic as a result of insulin resistance (Nishimura, 1969). From these findings, it seems likely that MF exerts its antidiabetic activity by decreasing insulin resistance. Many active compounds associated with antidiabetic activity have been discov-

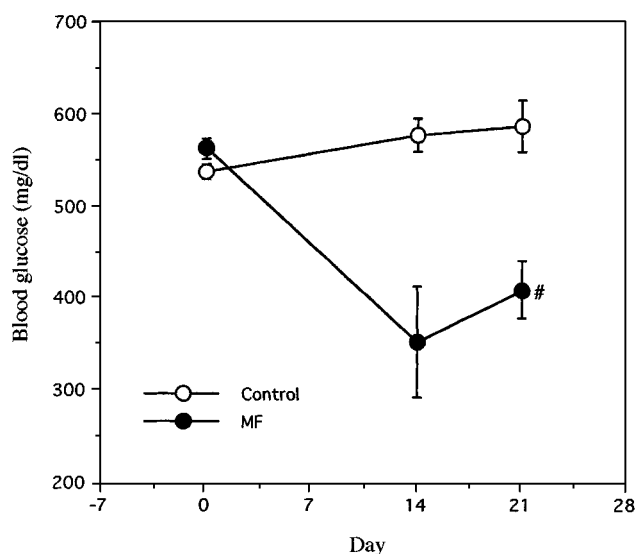


Fig. 2. Effect of MF on Blood Glucose in KK-Ay Mice. Each value represents the mean $\pm$ S.D. from 5 mice (significantly different from control, # $p < 0.01$).

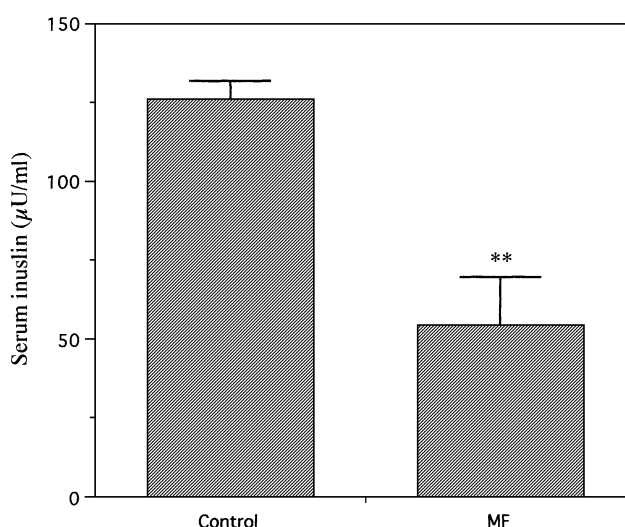


Fig. 3. Effects of MF on Insulin Concentration in KK-Ay Mice. Each value represents the mean $\pm$ S.D. from 5 mice (significantly different from control, ** $p < 0.01$).

ered, including triterpenoid glucoside (Kako et al., 1997), a steroidal glucoside (Kato et al., 1995), and an iridoidal glucoside (Miura et al., 1996). It is therefore interesting that xanthone compounds have a significant antidiabetic effect. Further study will indicate how MF could be useful in the clinical treatment of diabetes.

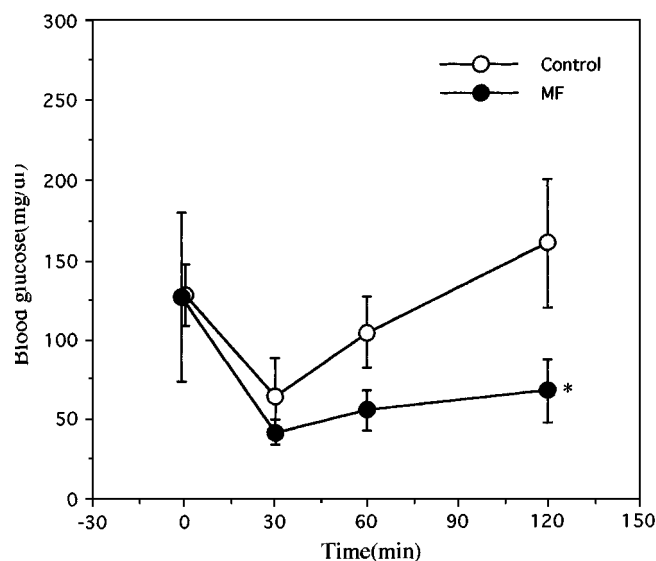


Fig. 4. Insulin-Tolerance Test of MF. Each value represents the mean $\pm$ S.D. from 5 mice (significantly different from control, * $p < 0.05$).

- Frahm, A. W., Chaudhuri, R. K.: ^{13}C -NMR spectroscopy of substituted xanthenes. *Tetrahedron* 35: 2035–2038, 1979.
- Fujita, M., Inoue, T.: Studies on the Constituents of *Iris florentina* L. C-glucosides of Xanthenes and Flavones from the Leaves. *Chem. Pharm. Bull.* 30: 2342–2348, 1982.
- Ichiki, H., Miura, T., Kubo, M., Ishihara, E., Komatsu, Y., Okada, M., Tanigawa, K., New Antidiabetic Compounds, mangiferin and its Glucoside. *Biol. Pharm. Bull.* 21: 1389–1390, 1998.
- Kako, M., Miura, T., Nishiyama Y., Ichimaru, M., Moriyasu, M., Kato, A.: Hypoglycemic Activity of Some Triterpenoid Glycosides. *J. Nat. Prod.* 60: 604–605, 1997.
- Kato, A., Miura, T., Fukunaga, T., Effects of steroidal Glycosides on Blood Glucose in Normal and Diabetic Mice. *Biol. Pharm. Bull.* 18: 167–168, 1995.
- Miura, T., Nishiyama Y., Ichimaru, M., Moriyasu, M., Kato, A.: Hypoglycemic Activity and structure-Activity Relationship of Iridoidal Glycosides. *Biol. Pharm. Bull.* 19: 160–161, 1996.
- Nishimura, M.: Breeding of mice strains for diabetes mellitus. *Exp. Animal* 18: 147–157, 1969.
- Stevens, J. F.: Determination of blood glucose by glucose oxidase method. *Clin. Chim. Acta* 32: 199, 1951.

References

- Bailei, C. J., Ahmed-Sorour, H.: Role of ovarian hormones in the long-term control of glucose homeostasis. *Diabetologia* 19: 475, 1980.

Address

- T. Miura, Suzuka University of Medical Science, 1001-1 Kishioka-cho, Suzuka, Mie 510-0293, Japan.